Richie Mouhouadi

Data & AI Engineer — Python · MLOps · CI/CD · Elasticsearch · AWS

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French & English · Available as Data & AI Engineer from March 2027 (end of MSc & apprenticeship)

PROFILE

Former R&D reservoir engineer turned Data & AI engineer. I take data use cases from exploration to production: data preparation, modelling, APIs, containerisation, CI/CD and monitoring. At SUEZ I work alongside operational and product teams to industrialise data products on production databases; on my own I ship and run full-stack systems with tests, CI pipelines, access control and secrets handling. Looking to join an agile squad building and running advanced data & AI use cases at scale.

EXPERIENCE

SUEZ Digital Solutions — Smart Metering Apprentice (Data & Project) *Paris · Feb 2025 – Mar 2027*

Prepaid smart water solution for emerging markets (Africa & Latin America)

- Support the project manager on client contract delivery (Build & Run) for pilots in Senegal (5,000 meters, scaling toward national rollout) and Luanda, Angola (5,000 meters).
- **Data industrialisation:** extracted undocumented LoRaWAN telemetry (RSSI, SNR, gateways) from JSON fields in production MySQL databases, validated field semantics with the product manager, and built the ingestion (JSON unpacking, timestamp normalisation, relationship model) behind a network-health supervision dashboard in Power BI.
- **MLOps:** designing a predictive-maintenance pipeline for metering assets (gateways, transmitters, meters) — Airflow orchestration, FastAPI serving, model tracking and drift monitoring — with a governance design that keeps production data off personal infrastructure.
- **Analytics for business:** building a study on 10k+ households and ~3 years of prepaid data to support tenders: adoption, recharge behaviour, consumption profiles and socio-economic drivers.

Beicip-Franlab — Reservoir Engineer Intern (Machine Learning) *Rueil-Malmaison · Jan – Jun 2024*

- Trained a GAN on fewer than 130 images to reproduce stratigraphic sequences from a Monte Carlo simulation, replacing slow simulation runs with fast generation.
- Forecast oil production with an LSTM. A graph neural network for material balance underperformed; I traced it to missing physical inputs (inter-well porosity, flow mechanisms) and documented what the next model needed.

Chang'an University — R&D Reservoir Engineer *Xi'an, China · Sep 2020 – Jun 2023*

- Contributed to 3 scientific papers and 1 patent application — ORCID: 0009-0007-2470-2110

PROJECTS — SOLO, FROM ZERO

DevOps Portfolio — CI/CD pipelines, containers & cloud deployment *github.com/rdmouhouadi/DevOps-Portfolio*

- **CI/CD with Jenkins & GitLab:** Freestyle, Pipeline and Multibranch jobs (Jenkinsfile) building a Maven artifact into a Docker image pushed to a private registry; pipelines auto-triggered by GitLab webhooks with dynamic versioning.
- **Continuous deployment to AWS EC2** with Docker Compose from the Jenkins pipeline; Nexus as artifact repository and private Docker registry (roles, volumes); secured Linux server and app deployment on DigitalOcean.

DocKA — Open-source document ingestion & search · v0.1.0 *github.com/rdmouhouadi/Docka*

- Python / FastAPI service that ingests PDF, Word, HTML and text into an **Elasticsearch** knowledge base (BM25), with PostgreSQL metadata, **Airflow** ingestion pipelines, Docker and unit tests.
- Roadmap to hybrid (semantic) search and evaluated RAG; data stays on the user's own infrastructure.

BuilderLab — Team-delivery platform for builders & learners · live beta *builderlab-tau.vercel.app*

- Next.js, TypeScript, Supabase (PostgreSQL + Row Level Security), Docker, GitHub Actions, Vercel. Authorisation enforced in the database (RLS policies, triggers), service-role key kept server-side.
- Code-reviewed by DSTI's scientific department before publication on the DSTI TechBlog; closed the findings: added CI, lint errors 54 → 0, component refactoring, containerised build.

StakPuls — Infrastructure alerting for solo builders · v0.6, on hold *stakpuls.com*

- LLM parsing of cloud alert emails into a single monitoring dashboard (TypeScript, Neon Postgres, Cloudflare Workers, Groq).

EDUCATION

MSc Data Science & AI — DSTI School of Engineering *Paris · Feb 2025 – Mar 2027*
DevOps Bootcamp — TechWorld with Nana · Kubernetes & Terraform next *Online · Aug 2025 – present*
MSc Geoenergies — University of Pau and the Adour Region *Pau · Sep 2023 – Jun 2024*

SKILLS

Data & ML	Python (pandas, NumPy, scikit-learn, matplotlib), SQL (MySQL, PostgreSQL), deep learning (GAN, LSTM, GNN), Power BI
MLOps / Data eng.	Airflow, FastAPI, model monitoring & drift, Elasticsearch, Spark (DSTI coursework)
DevOps & Cloud	Docker & Docker Compose, Jenkins, GitLab, GitHub Actions, Nexus, AWS (EC2), Linux, Git
Web & tooling	TypeScript, React / Next.js, Supabase, Vercel; Neo4j (certified)
Ways of working	Build & Run delivery with business and product teams, code review, documentation